

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 2

June 2024 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

27 adaptive question sections 104 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each question section appears in the same order as the Student book: the complete source demand is followed by its demand-local worked solution and final answer.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. For example, drawing boundary lines supports the Unit 2 region-of-inequalities demand. This book contains the demands owned by Unit 2; the selection contains 104 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Source assets are retained for private teaching use.

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Question 01

4MA1-1H Q1

ORIGINAL QUESTION

3 marks

- 1 Here are the first four terms of an arithmetic sequence.

1 4 7 10

- (a) Find an expression, in terms of n , for the n th term of this sequence.

.....
(2)

The n th term of a different arithmetic sequence is $5n + 17$

- (b) Find the 12th term of this sequence.

.....
(1)

Worked Solution 01

Arithmetic sequences

MODULAR UNIT 2

3 marks

General term

$$1 + 3(n - 1) = 3n - 2.$$

Twelfth term

$$5(12) + 17 = 77.$$

Final answer

$$3n - 2; \quad 77$$

Question 02

4MA1-1H Q3

ORIGINAL QUESTION

2 marks

- 3 Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

Worked Solution 02

Highest common factor

MODULAR UNIT 2

2 marks

Prime factors

$$72 = 2^3 \cdot 3^2, \quad 108 = 2^2 \cdot 3^3.$$

Common powers

$$2^2 \cdot 3^2 = 36.$$

Final answer

$$36$$

Question 03

4MA1-1H Q4

ORIGINAL QUESTION

3 marks

- 4 Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.

This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

..... kilometres

Worked Solution 03

Reverse percentage

MODULAR UNIT 2

3 marks

Reverse the increase

$$1.15m = 943 \Rightarrow m = 943 \div 1.15 = 820.$$

Final answer

820 km

Question 04

4MA1-1H Q5

ORIGINAL QUESTION

4 marks

- 5 In the diagram, $ABCDE$ is a regular pentagon.

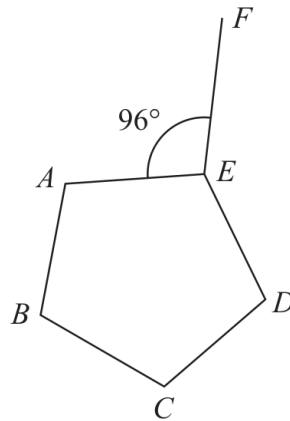


Diagram **NOT**
accurately drawn

Angle $AEF = 96^\circ$

Work out the size of the obtuse angle FED
Show your working clearly.

Worked Solution 04

Regular polygon angle

MODULAR UNIT 2

4 marks

Interior angle

$$(5 - 2)180^\circ / 5 = 108^\circ.$$

Around the point

$$\angle FED = 360^\circ - 96^\circ - 108^\circ = 156^\circ.$$

Final answer

$$156^\circ$$

Question 05

4MA1-1H Q9

ORIGINAL QUESTION

3 marks

- 9 The table gives the amount of rice produced by each of two countries in 2020

Country	Amount of rice (tonnes)
Indonesia	3.5×10^7
Argentina	8.2×10^5

- (a) Write 3.5×10^7 as an ordinary number.

.....
(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

- (b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

..... tonnes
(2)

Worked Solution 05

Standard form

MODULAR UNIT 2

3 marks

Ordinary number

$$3.5 \times 10^7 = 35\,000\,000.$$

Addition

$$8.2 \times 10^5 + 6\,780\,000 = 7\,600\,000 = 7.6 \times 10^6.$$

Final answer

$$35\,000\,000; \quad 7.6 \times 10^6 \text{ tonnes}$$

Question 06

4MA1-1H Q14

ORIGINAL QUESTION

5 marks

14 The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

..... dinars

Worked Solution 06

Linked ratio

MODULAR UNIT 2**5 marks****Abel and Bahira**

$$B = 1.45A, \quad A + B = 15435 \Rightarrow A = 6300, \quad B = 9135.$$

Chanda

$$B = \frac{3}{2}C \Rightarrow C = \frac{2}{3}(9135) = 6090.$$

Final answer

6090 dinars

Question 07

4MA1-1H Q15

ORIGINAL QUESTION

4 marks

15 The function f is defined as

$$f: x \mapsto \frac{3x + 1}{x - 2}$$

(a) State the value of x that cannot be included in any domain of the function f

.....
(1)

(b) Express the inverse function f^{-1} in the form $f^{-1}(x) = \dots$

$f^{-1}(x) = \dots$
(3)

Worked Solution 07

Domain and inverse function

MODULAR UNIT 2

4 marks

Domain

$$x - 2 \neq 0 \Rightarrow x \neq 2.$$

Inverse

$$y = (3x + 1)/(x - 2) \Rightarrow x(y - 3) = 2y + 1 \Rightarrow f^{-1}(x) = \frac{2x + 1}{x - 3}.$$

Final answer

$$x \neq 2; \quad f^{-1}(x) = \frac{2x + 1}{x - 3}$$

Question 08

4MA1-1H Q18

ORIGINAL QUESTION

5 marks

18 A curve C has equation $y = x^3 - 40x + 1$

Find the coordinates of both the points on C at which the gradient is 8

(..... ,)

(..... ,)

Worked Solution 08

Gradient on a curve

MODULAR UNIT 2

5 marks

Differentiate

$$dy/dx = 3x^2 - 40.$$

Set the gradient

$$3x^2 - 40 = 8 \Rightarrow x = \pm 4.$$

Coordinates

$$y(4) = -95, \quad y(-4) = 97.$$

Final answer

$$(4, -95), (-4, 97)$$

Question 09

4MA1-1H Q21

ORIGINAL QUESTION

2 marks

21 A curve has equation $y = f(x)$

There is one minimum point on this curve.

The coordinates of this minimum point are $(5, -4)$

Write down the coordinates of the minimum point on the curve with equation

(i) $y = f(x + 7)$

(..... ,)
(1)

(ii) $y = f(x) - 6$

(..... ,)
(1)

Worked Solution 09

Graph transformations

MODULAR UNIT 2

2 marks

Horizontal shift

$$y = f(x + 7) : (5, -4) \mapsto (-2, -4).$$

Vertical shift

$$y = f(x) - 6 : (5, -4) \mapsto (5, -10).$$

Final answer

$$(-2, -4); \quad (5, -10)$$

Question 10

4MA1-1H Q23

ORIGINAL QUESTION

5 marks

- 23 A solid shape is made by removing a hemisphere, shown shaded, from a cone as shown in the diagram.

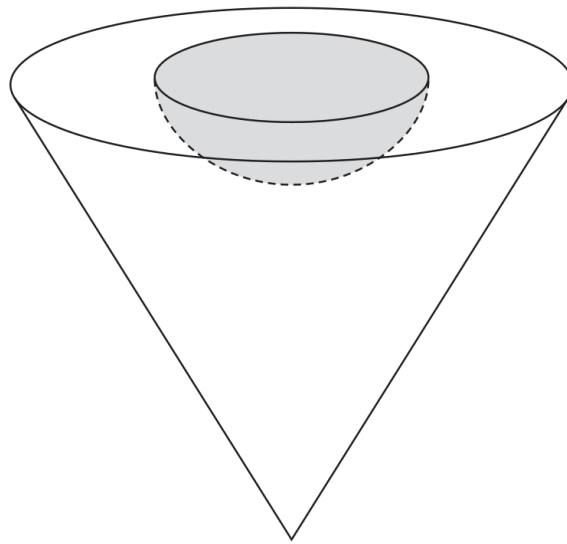


Diagram **NOT**
accurately drawn

The radius of the hemisphere is $2x$ cm

The radius of the base of the cone is $5x$ cm

The vertical height of the cone is $6x$ cm

The volume of the solid shape is 6948π cm³

Work out the **total** surface area of the solid hemisphere that has been removed from the cone.

Give your answer correct to the nearest integer.

..... cm²

Worked Solution 10

Cone with a hemisphere removed

MODULAR UNIT 2

5 marks

Find the scale

$$50\pi x^3 - \frac{16}{3}\pi x^3 = 6948\pi \Rightarrow x^3 = 10422/67.$$

Removed surface

$$3\pi(2x)^2 = 12\pi x^2 = 12\pi(10422/67)^{2/3} = 1090.390\dots$$

Final answer

$$1090 \text{ cm}^2$$

Question 11

4MA1-1H Q24

ORIGINAL QUESTION

6 marks

24 A polygon has n sides, where $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is 84°

The common difference of the sequence is 4°

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

Worked Solution 11

Arithmetic sequence of polygon angles

MODULAR UNIT 2

6 marks

Two sums

$$S_n = \frac{n}{2}[168 + 4(n - 1)] = 2n^2 + 82n, \quad S_n = 180(n - 2).$$

Solve

$$n^2 - 49n + 180 = 0 \Rightarrow n = 45 \ (n > 5).$$

Angle sum

$$180(45 - 2) = 7740.$$

Final answer

$$7740^\circ$$

Question 12

4MA1-2H Q1

ORIGINAL QUESTION

3 marks

- 1 Here are eight numbers written in order of size

$$h \quad 6 \quad 7 \quad 8 \quad j \quad 16 \quad k \quad k$$

where h, j and k are integers.

The median of the eight numbers is 10

The mode of the eight numbers is 18

The range of the eight numbers is 13

Work out the value of h , the value of j and the value of k

 $h = \dots\dots\dots$ $j = \dots\dots\dots$ $k = \dots\dots\dots$

Worked Solution 12

Median, mode and range

MODULAR UNIT 2

3 marks

Median

$$(8 + j)/2 = 10 \Rightarrow j = 12.$$

Mode and range

$$k = 18, \quad k - h = 13 \Rightarrow h = 5.$$

Final answer

$$h = 5, \quad j = 12, \quad k = 18$$

Question 13

4MA1-2H Q2

ORIGINAL QUESTION**4 marks**

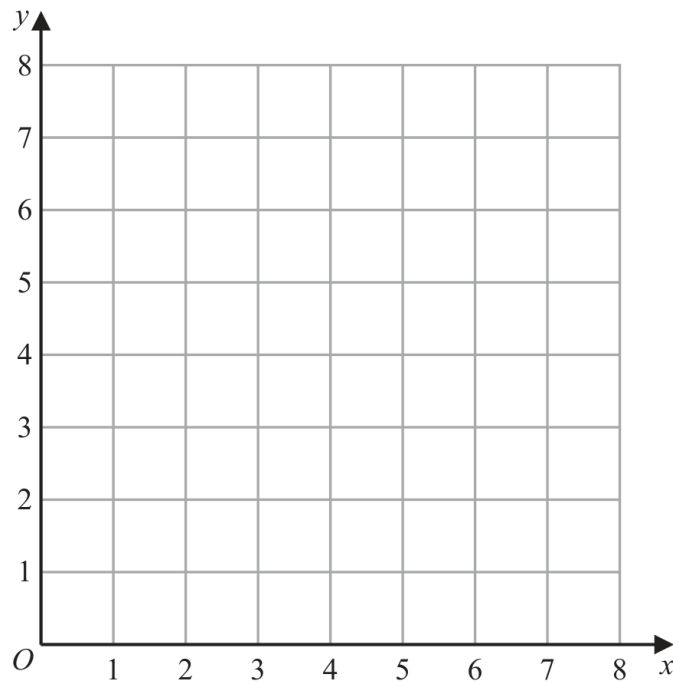
2 (a) On the grid, draw the straight line with equation

(i) $y = 2$

(ii) $x = 6$

(iii) $y = x + 1$

Label each line with its equation.



(3)

(b) Show, by shading on the grid, the region that satisfies all three of the inequalities

$$y \geq 2$$

$$x \leq 6$$

$$y \leq x + 1$$

Label the region **R**

(1)

Worked Solution 13

Lines and a region of inequalities

MODULAR UNIT 2

4 marks

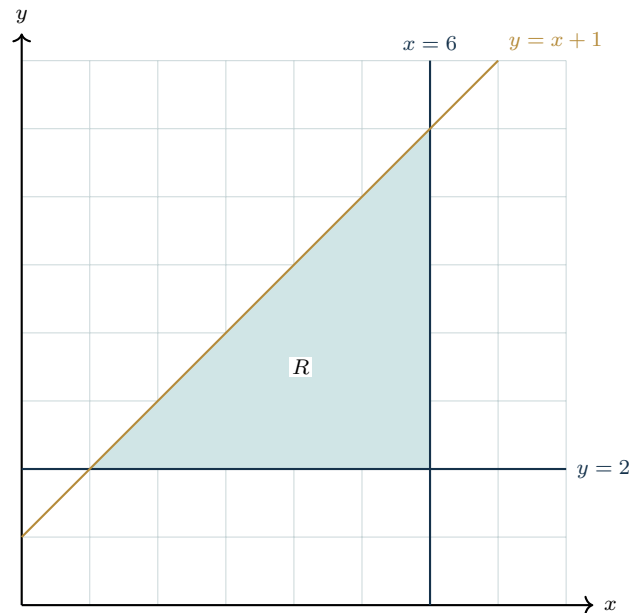
BoundariesDraw and label $y = 2$, $x = 6$, $y = x + 1$.**Region**Shade the closed triangle with vertices $(1, 2)$, $(6, 2)$, $(6, 7)$ and label R .**Final answer**The triangular region R

Completed diagram 13

Lines and a region of inequalities

MODULAR UNIT 2

4 marks

Completed solution diagram: Lines and feasible region

Question 14

4MA1-2H Q8

ORIGINAL QUESTION

5 marks

- 8 Ishir plants 600 bulbs in a garden.
He plants tulip bulbs, crocus bulbs and daffodil bulbs so that

number of tulip bulbs : number of crocus bulbs : number of daffodil bulbs = 9 : 4 : 2

45% of the tulip bulbs are for yellow flowers.

$\frac{5}{8}$ of the crocus bulbs are for yellow flowers.

All of the daffodil bulbs are for yellow flowers.

Work out the number of bulbs that are for yellow flowers.

Worked Solution 14

Ratio and percentages

MODULAR UNIT 2

5 marks

Bulb counts

$$600/15 = 40; \quad T = 360, C = 160, D = 80.$$

Yellow

$$0.45(360) + \frac{5}{8}(160) + 80 = 342.$$

Final answer

342 bulbs

Question 15

4MA1-2H Q9

ORIGINAL QUESTION

3 marks

- 9 Giovanni invests 4500 koruna in a savings account for 4 years.
He gets 2.4% per year compound interest.

Work out how much money Giovanni will have in the savings account at the end of 4 years.

Give your answer correct to the nearest koruna.

..... koruna

Worked Solution 15

Compound interest

MODULAR UNIT 2

3 marks

Multiplier

$$4500(1.024)^4 = 4947.802 \dots$$

Final answer

4948 koruna

Question 16

4MA1-2H Q10

ORIGINAL QUESTION

3 marks

10 Solve the simultaneous equations

$$\begin{aligned}6x + 4y &= 1 \\ 3x + 5y &= 8\end{aligned}$$

Show clear algebraic working.

$x = \dots\dots\dots$

$y = \dots\dots\dots$

Worked Solution 16

Simultaneous equations

MODULAR UNIT 2

3 marks

Eliminate

$$2(3x + 5y = 8) \Rightarrow 6x + 10y = 16; \quad 6y = 15.$$

Finish

$$y = 2.5, \quad x = -1.5.$$

Final answer

$$x = -1.5, \quad y = 2.5$$

Question 17

4MA1-2H Q12

ORIGINAL QUESTION

3 marks

12 Ali uses a fitness tracker to count the number of steps he walks each day for 7 days.

For the first 4 days, his mean number of steps is 11 800

For the next 3 days, his mean number of steps is 13 207

Work out his mean number of steps for the 7 days.

Worked Solution 17

Combined mean

MODULAR UNIT 2

3 marks

Totals

$$4(11800) = 47200, \quad 3(13207) = 39621.$$

Mean

$$(47200 + 39621)/7 = 12403.$$

Final answer

12403 steps

Question 18

4MA1-2H Q13

ORIGINAL QUESTION**7 marks**

13 The table gives information about the distances, in km, that 70 teachers travel to school.

Distance (d km)	Frequency
$0 < d \leq 10$	7
$10 < d \leq 20$	17
$20 < d \leq 30$	18
$30 < d \leq 40$	14
$40 < d \leq 50$	10
$50 < d \leq 60$	4

(a) Complete the cumulative frequency table.

Distance (d km)	Cumulative frequency
$0 < d \leq 10$	
$0 < d \leq 20$	
$0 < d \leq 30$	
$0 < d \leq 40$	
$0 < d \leq 50$	
$0 < d \leq 60$	

(1)

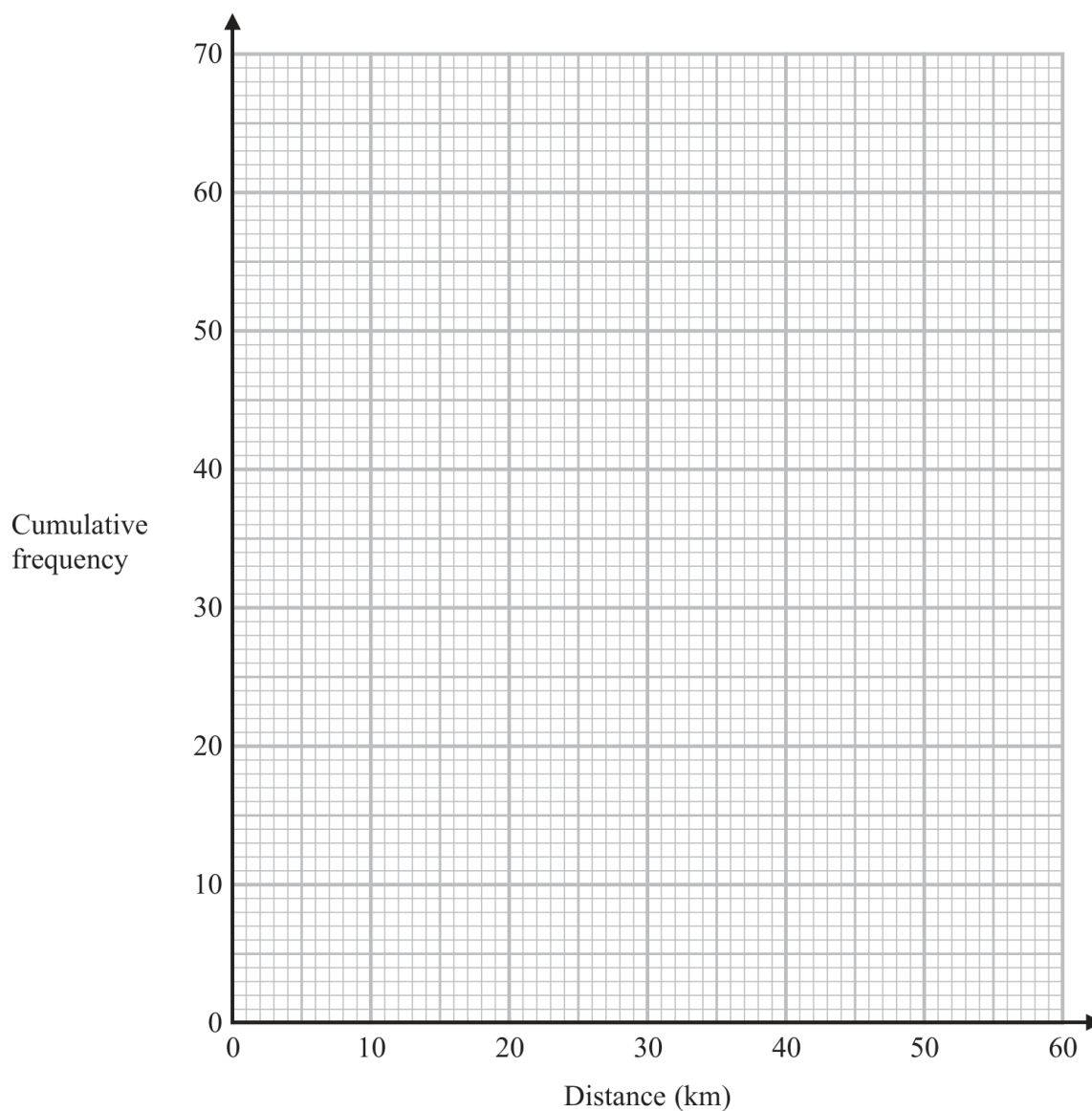
Question 18 (continued)

4MA1-2H Q13

ORIGINAL QUESTION**7 marks**

(b) On the grid opposite, draw a cumulative frequency graph for your table.

(2)



(c) Use your graph to find an estimate for the interquartile range of the distances.

..... km

(2)

Question 18 (continued)

4MA1-2H Q13

ORIGINAL QUESTION

7 marks

- (d) Use your graph to find an estimate for the number of teachers who travel more than 46 km.

.....

(2)

Worked Solution 18

Cumulative frequency

MODULAR UNIT 2

7 marks

Table $7, 24, 42, 56, 66, 70.$ **Graph**Plot $(10, 7), (20, 24), (30, 42), (40, 56), (50, 66), (60, 70)$ and join smoothly.**IQR**

$$Q_1 \approx 16.2, Q_3 \approx 37.5 \Rightarrow \text{IQR} \approx 21.3 \text{ km.}$$

More than 46 km

$$70 - \text{CF}(46) \approx 70 - 62 = 8.$$

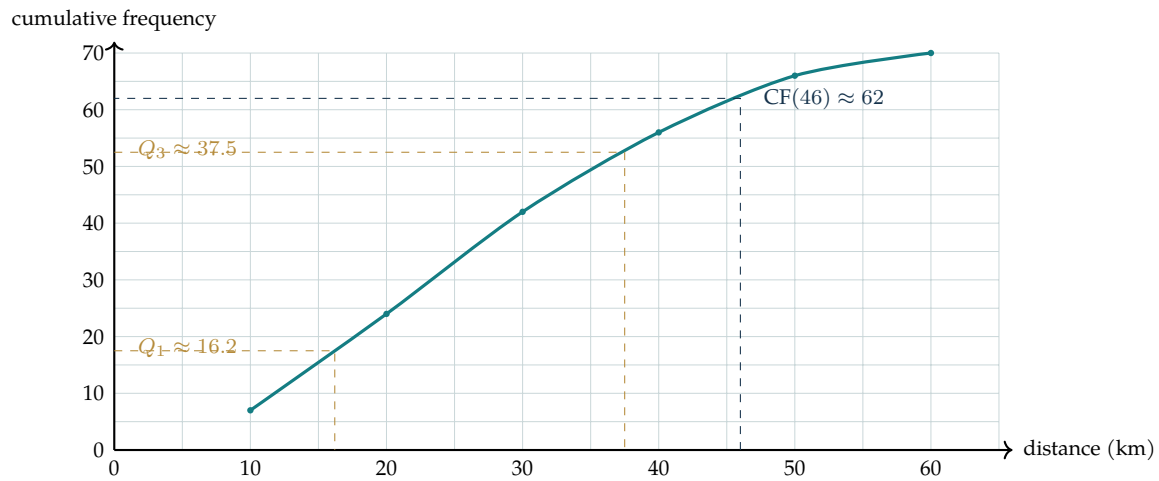
Final answerCF 7, 24, 42, 56, 66, 70; IQR ≈ 21.3 km; 8 teachers

Completed diagram 18

Cumulative frequency

MODULAR UNIT 2

7 marks

Completed solution diagram: Cumulative-frequency graph

Question 19

4MA1-2H Q14

ORIGINAL QUESTION

3 marks

14 (a) Show that $3y(2y + 5)(y + 7)$ can be written in the form $ay^3 + by^2 + cy$ where a, b and c are integers.

(3)

Worked Solution 19

Expansion

MODULAR UNIT 2

3 marks

Expand the two brackets

$$(2y + 5)(y + 7) = 2y^2 + 19y + 35.$$

Multiply by the outside factor

$$3y(2y^2 + 19y + 35) = 6y^3 + 57y^2 + 105y.$$

Final answer

$$6y^3 + 57y^2 + 105y$$

Question 20

4MA1-2H Q15

ORIGINAL QUESTION

4 marks

15 (a) Make g the subject of $e = \sqrt{\frac{7g+5}{11+2g}}$

(4)

Worked Solution 20

Change the subject

MODULAR UNIT 2

4 marks

Square and remove the fraction

$$e^2(11 + 2g) = 7g + 5.$$

Collect the terms in the new subject

$$g(2e^2 - 7) = 5 - 11e^2 \Rightarrow g = \frac{5 - 11e^2}{2e^2 - 7}.$$

Final answer

$$g = \frac{5 - 11e^2}{2e^2 - 7}$$

Question 21

4MA1-2H Q15

ORIGINAL QUESTION

3 marks

- (b) Solve the inequality $3y^2 + 4y - 32 > 0$
Show your working clearly.

.....

(3)

Worked Solution 21

Quadratic inequality

MODULAR UNIT 2

3 marks

Factorise

$$3y^2 + 4y - 32 = (3y - 8)(y + 4).$$

Use the outside intervals

$$(3y - 8)(y + 4) > 0 \Rightarrow y < -4 \text{ or } y > \frac{8}{3}.$$

Final answer

$$y < -4 \text{ or } y > \frac{8}{3}$$

Question 22

4MA1-2H Q17

ORIGINAL QUESTION

3 marks

17 Q is directly proportional to the square root of d

$$Q = 4.5 \text{ when } d = 324$$

Find a formula for Q in terms of d

Worked Solution 22

Direct proportion

MODULAR UNIT 2

3 marks

Constant

$$Q = k\sqrt{d}, \quad 4.5 = 18k \Rightarrow k = 1/4.$$

Final answer

$$Q = \frac{\sqrt{d}}{4}$$

Question 23

4MA1-2H Q20

ORIGINAL QUESTION

3 marks

20 Given that $k = x - y$ and $x = \frac{1}{4y}$

express $\frac{5k}{x+2}$ in the form $\frac{a-by^2}{c+dy}$ where a, b, c and d are integers.

.....

Worked Solution 23

Algebraic substitution

MODULAR UNIT 2

3 marks

Substitute

$$x = 1/(4y), \quad k = 1/(4y) - y.$$

Simplify

$$\frac{5k}{x+2} = \frac{5-20y^2}{1+8y}.$$

Final answer

$$\frac{5-20y^2}{1+8y}$$

Question 24

4MA1-2H Q21

ORIGINAL QUESTION

5 marks

- 21 The diagram shows a square $ABCD$ and a circle.

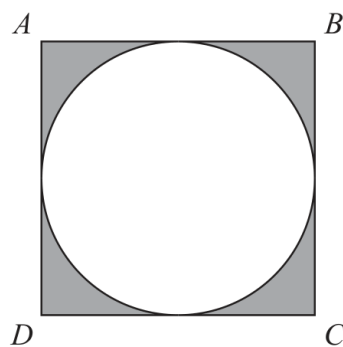


Diagram **NOT**
accurately drawn

The sides of the square are tangents to the circle.

The total area of the shaded regions is 80 cm^2

Work out the length of AC

Give your answer correct to 3 significant figures.

..... cm

Worked Solution 24

Square and inscribed circle

MODULAR UNIT 2

5 marks

Area

$$(2r)^2 - \pi r^2 = 80 \Rightarrow r^2 = 80/(4 - \pi).$$

Diagonal

$$AC = 2r\sqrt{2} = 27.305 \dots$$

Final answer

27.3 cm

Question 25

4MA1-2H Q22

ORIGINAL QUESTION

5 marks

22 The straight line **L** has equation $x + y = 5$

The curve **C** has equation $2x^2 + 3y^2 = 210$

Find the coordinates of the points where **L** and **C** intersect.
Show clear algebraic working.

(.....,) (.....,)

Worked Solution 25

Line-curve intersections

MODULAR UNIT 2

5 marks

Substitute

$$y = 5 - x : \quad 2x^2 + 3(5 - x)^2 = 210.$$

Solve

$$x^2 - 6x - 27 = 0 \Rightarrow x = 9, -3.$$

Coordinates

$$y = -4, 8.$$

Final answer

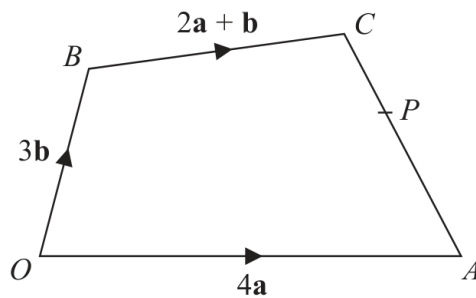
$$(9, -4), (-3, 8)$$

Question 26

4MA1-2H Q24

ORIGINAL QUESTION**6 marks**

24

Diagram **NOT**
accurately drawnThe diagram shows a quadrilateral $OACB$ in which

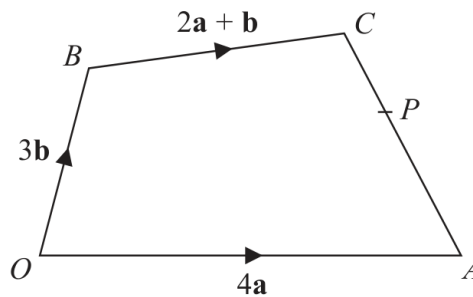
$$\vec{OA} = 4\mathbf{a} \quad \vec{OB} = 3\mathbf{b} \quad \vec{BC} = 2\mathbf{a} + \mathbf{b}$$

- (a) Find $\vec{AC}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.

$$\vec{AC} = \dots\dots\dots (2)$$

Question 26 (continued)

4MA1-2H Q24

ORIGINAL QUESTION**6 marks****24**Diagram **NOT**
accurately drawn

The diagram shows a quadrilateral $OACB$ in which

$$\overrightarrow{OA} = 4\mathbf{a} \quad \overrightarrow{OB} = 3\mathbf{b} \quad \overrightarrow{BC} = 2\mathbf{a} + \mathbf{b}$$

The point P lies on AC such that $AP:PC = 3:2$

The point Q is such that OPQ and BCQ are straight lines.

- (b) Using a vector method, find $\overrightarrow{OQ}$ in terms of $\mathbf{a}$ and $\mathbf{b}$
Give your answer in its simplest form.
Show your working clearly.

$$\overrightarrow{OQ} = \dots\dots\dots (4)$$

Worked Solution 26

Vectors and collinearity

MODULAR UNIT 2

6 marks

Part (a)

$$\vec{AC} = 4b - 2a.$$

Point P

$$\vec{OP} = 4a + \frac{3}{5}(4b - 2a) = \frac{14}{5}a + \frac{12}{5}b.$$

Point Q

$$\vec{OQ} = \lambda \vec{OP} = 3b + \mu(2a + b) \Rightarrow \lambda = 3.$$

Final answer

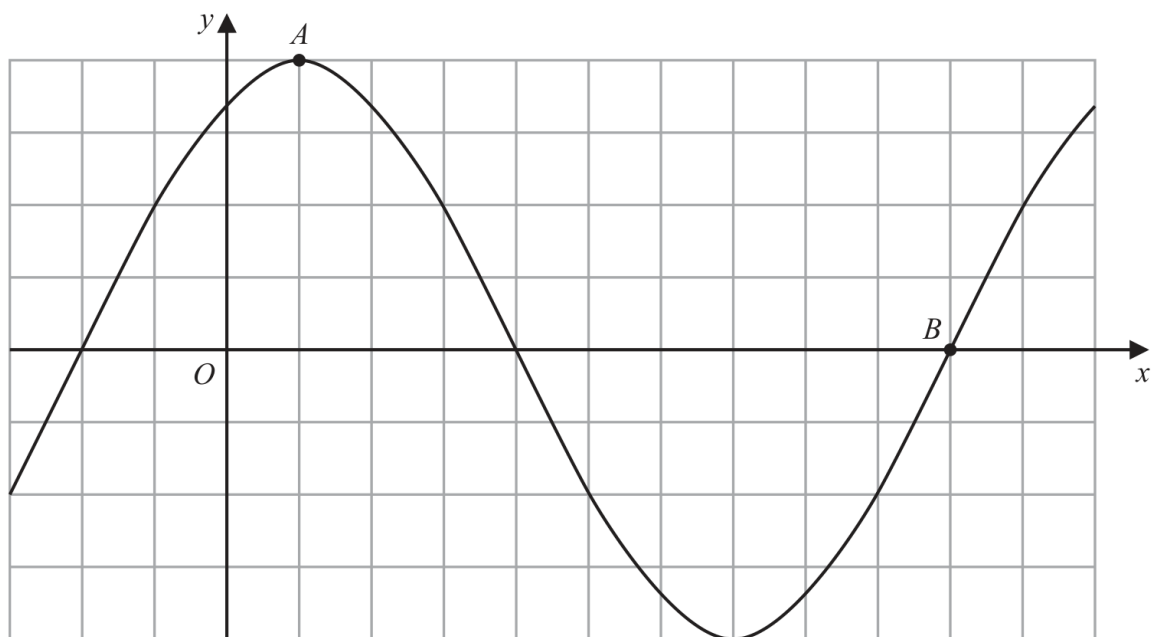
$$\vec{OQ} = \frac{42}{5}a + \frac{36}{5}b$$

Question 27

4MA1-2H Q25

ORIGINAL QUESTION**2 marks**

25 The diagram shows a sketch of the graph of $y = 2\sin(x + 60)^\circ$



(i) Find the coordinates of the point A

(.....,)
(1)

(ii) Find the coordinates of the point B

(.....,)
(1)

Worked Solution 27

Transformed sine graph

MODULAR UNIT 2

2 marks

Maximum

$$x + 60 = 90 \Rightarrow A = (30, 2).$$

Axis crossing

$$x + 60 = 360 \Rightarrow B = (300, 0).$$

Final answer

$$A = (30, 2), \quad B = (300, 0)$$